

SUPPLEMENTARY MATERIAL

Physiologically Based Pharmacokinetic Modeling to Assess Transporter-mediated Drug–Drug Interactions in Support of Labeling Recommendations for Zavegepant Nasal Spray, an Acute Treatment for Migraine

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Supplementary Table S1. Trial design for Simcyp simulations in healthy participants

N individuals × N trials	Zavegepant dose	Zavegepant treatment	Precipitant drug and dose	Precipitant treatment	Analysis	Ref
Model development						
18 × 10 ^a	100 mg oral	Day 1	None	None	Pred vs Obs	Bhardwaj R. et al ¹
18 × 10 ^a	50 mg oral	Day 8	MD itraconazole 200 mg QD ^c	Days 1–9	Pred vs Obs	Bhardwaj R. et al ¹
15 × 10 ^a	100 mg oral	Day 10	MD rifampin 600 mg QD ^c	Days 1–10	Pred vs Obs	Bhardwaj R. et al ¹
9 × 10 ^a	20 mg IN	Days 1–14	None	None	Pred vs Obs	Bertz R. et al. ²
Model Validation						
9–12 × 10 ^a	50, 100, 200, 300 mg oral	Days 1–10	None	None	Pred vs Obs	DoF
9 × 10 ^a	5, 10 mg IN	Days 1–14	None	None	Pred vs Obs	Bertz R. et al. ²
18 × 10 ^a	10 mg IN	Day 4	MD itraconazole 200 mg QD ^c	Days 1–9	Pred vs Obs	Bhardwaj R. et al ¹
Model application ^c						
10 × 10 ^b	10 mg IN	Day 1	SD rifampin 600 mg ^c	Day 1	Pred	NA
10 × 10 ^b	10 mg IN	Day 10	MD rifampin 600 mg QD ^c	Days 1–10	Pred	NA
10 × 10 ^b	10 mg IN	Day 1	SD cyclosporine A 600 mg ^c	Day 1	Pred	NA
10 × 10 ^b	10 mg IN	Day 10	MD carbamazepine 300 mg BID ^c	Days 1–11	Pred	NA
10 × 10 ^b	10 mg IN	Day 10	MD efavirenz 600 mg QD ^c	Days 1–11	Pred	NA

^aSimcyp parameters for the simulated population matched age range and % female in the associated clinical study.

^bModel application used default age range (20–50 years) and 50% female to match the clinical drug–drug interaction study.

^cAlso included a simulation of zavegepant administered alone on Day 1 (i.e., in the absence of precipitant drug).

DoF, data on file; HP, healthy participant; IN, intranasal; MD, multiple dose; NA, not applicable; N, number; Obs, observed; Pred, predicted; QD, once daily; Ref, study reference; SD, single dose.

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1. Bhardwaj R. *et al.* Mass balance and pharmacokinetic characterization of zavegepant in healthy male subjects. *Clin Transl Sci.* **17**, e70015 (2024).
2. Bertz R. *et al.* Safety, tolerability, and pharmacokinetics of single and multiple ascending doses of zavegepant nasal spray in healthy adults from two phase 1 randomized, placebo-controlled trials. *Headache.* **65**, 1331-1343 (2025).

Supplementary Table S2. Simcyp PBPK model input parameters and source references for rifampin multiple-dose (SV-Rifampin-MD)

Parameter	Value	Method/Ref
Molecular weight (g/mol)	823	PubChem
LogP	4.01	Stephen KW. et al ¹ Liu L. et al ²
Compound type	Ampholyte	
pK _a	1.7, 7.9	Pasqualucci CR. et al ³
B/P	0.9	Loos U. et al ⁴
<i>Fu</i>	0.116	Bowman G. et al ⁵ Templeton IE. et al ⁶
Main binding protein	Human serum albumin	Boman G. ⁷
<i>Fa</i>	0.93	Predicted
<i>Fu_{gut}</i>	1	Assumed
<i>Q_{gut}</i> (L/h)	9.42	Predicted from scaling Caco-2 data
<i>P_{eff,man}</i> (10 ⁻⁴ cm/s)	2.15	Predicted from scaling Caco-2 data
Distribution model	Minimal PBPK model	File input
<i>V_{ss}</i> (L/kg)	0.42	Acocella G. et al ⁸ Acocella G. et al ⁹ Acocella G. et al ¹⁰ Acocella G. et al ¹¹ Nitti V. et al ¹² Lau Y. et al ¹³
<i>CL_{IV}</i> (L/h)	8.7	Acocella G. et al ¹¹
<i>CL_R</i> (L/h)	1.26	Acocella G. ¹⁴ Peloquin CA. et al ¹⁵ Polk RE. et al ¹⁶
CYP1A2 <i>Ind_{max}</i>	2.7	Dixit V. et al ¹⁷
CYP1A2 <i>Ind_{C50}</i> (uM)	0.1	Optimized to recover clinically reported interaction with theophylline (Gillum JG. et al ¹⁸)
CYP2B6 <i>Ind_{max}</i>	5.04	In-house data Dixit V. et al ¹⁹
CYP2B6 <i>Ind_{C50}</i> (uM)	0.07	In-house data
CYP2C8 <i>K_i</i> (uM)	24.5	Kajosaari LI. et al ²⁰
CYP2C8 <i>Ind_{max}</i>	6.7	Derived from in vitro mRNA induction data in human hepatocytes (Dixit V. et al ¹⁷)
CYP2C8 <i>Ind_{C50}</i> (uM)	0.3	
CYP2C9 <i>Ind_{max}</i>	6	Optimized to recover reported interaction with tolbutamide (Kirby BJ. et al ²¹)
CYP2C9 <i>Ind_{C50}</i> (uM)	0.1	Optimized
CYP3A4 <i>K_i</i> (uM)	15	Kajosaari LI. et al ²⁰
CYP3A4 <i>Ind_{max}</i>	16	Optimized (Almond LM. et al ²²)
CYP3A4 <i>Ind_{C50}</i> (uM)	0.32	Almond LM. et al ²²
CYP3A5 <i>Ind_{max}</i>	16	Optimized (Almond LM. et al ²²)
CYP3A5 <i>Ind_{C50}</i> (uM)	0.32	Almond LM. et al ²²
UGT1A1 <i>Ind_{max}</i>	3.16	Smith CM. et al ²³
UGT1A1 <i>Ind_{C50}</i> (uM)	0.39	Smith CM. et al ²³

B/P, blood-to-plasma ratio; *CL_{IV}*, intravenous clearance; *CL_R*, renal clearance; CYP, cytochrome P450; *Fa*, fraction absorbed; *Fu*, fraction unbound in plasma; *Fu_{gut}*, fraction unbound in gut; *Ind_{C50}*, concentration producing 50% of maximum induction; *Ind_{max}*, maximum induction effect; *K_i*, inhibition constant; LogP, octanol/water partition coefficient; MD, multiple dose; PBPK, physiologically based pharmacokinetic; *P_{eff,man}*, effective human jejunal

permeability; pKa, acid dissociation constant; Q_{gut} , gut blood flow term used in the absorption model; Ref, study reference; UGT, uridine diphosphate glucuronosyltransferase; V_{ss} , volume of distribution at steady state.

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1. Stephen KW. *et al.* Factors determining the passage of drugs from blood into saliva. *Br J Clin Pharmacol.* **9**, 51-55 (1980).
2. Liu L. *et al.* 2010. Radiosynthesis and bioimaging of the tuberculosis chemotherapeutics isoniazid, rifampicin and pyrazinamide in baboons. *J Med Chem.* **53**, 2882-2889 (2010).
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20. Kajosaari LI. *et al.* Metabolism of repaglinide by Cyp2c8 and Cyp3a4 in vitro: effect of fibrates and rifampicin. *Basic Clin Pharmacol Toxicol.* **97**, 249-256 (2005).
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Supplementary Table S3. Simcyp PBPK model input parameters and source references for rifampin single-dose (SV-Rifampin-SD)

Parameter	Value	Method/Ref
Molecular weight (g/mol)	823	PubChem
LogP	4.01	Liu L. et al ¹ Stephen KW. et al ²
Compound type	Ampholyte	
pK _a	1.7 (acid), 7.9 (base)	Pasqualucci CR. et al ³
B/P	0.9	Loos U. et al ⁴
<i>Fu</i>	0.116	Boman G. et al ⁵ Templeton IE. et al ⁶
Main plasma binding protein	Human serum albumin	Boman G. ⁷
<i>Fu_{gut}</i>	1	Default value
<i>P_{eff,man}</i> (10 ⁻⁴ cm/s)	2.15	Predicted
Permeability assay	Caco-2	
Apical:basolateral pH	6.4:7.4	
Activity	Passive	
<i>P_{appA:B}</i> (10 ⁻⁶ cm/s)	15	Optimized
Reference compound	Propranolol	
Reference compound value (10 ⁻⁶ cm/s)	21.15	Default value
Distribution model	Full PBPK model	
<i>V_{ss}</i> (L/kg)	0.42	Predicted - Method 2
<i>K_p</i> scalar	0.0976	Optimized
<i>CL_{int}</i> (HLM) (uL/min/mg)	2.84	Optimized from <i>CL_{IV}</i> data Acocella G. et al ⁸ Nitti V. et al ⁹ Lau Y. et al ¹⁰
<i>CL_{int}</i> (Bile) (uL/min/10 ⁶ cells)	0.288	
<i>CL_R</i> (L/h)	1.26	
		Acocella G. ¹¹ Peloquin CA. et al ¹² Polk RE. et al ¹³
Liver model	Permeability-limited liver model	
<i>CL_{PD}</i> (mL/min/10 ⁶ hepatocytes)	10 ⁻⁵	Scaled from predicted <i>P_{eff,man}</i>
<i>Fu_{iw}</i>	0.044	Optimized
<i>Fu_{ew}</i>	1	Predicted
OATP1B1 <i>J_{max}</i> (pmol/min/pmol transporter)	5.52	Calculated from OATP1B1 abundance
OATP1B1 <i>K_m</i> (uM)	0.5	Optimized
P-gp <i>K_i</i> (uM): Gut & Liver	4.3	Pedersen JM. et al ¹⁴
BCRP <i>K_i</i> (uM): Gut & Liver	12.54	Prueksaritanont T. et al ¹⁵ Rosuvastatin DDI uses value equal to OATP2B1 <i>K_i</i>
MRP4 <i>K_i</i> (uM)	12.54	Brake LHM. et al ¹⁶
OATP1B1 <i>K_i</i> (uM)	0.168	Pahwa S. et al ¹⁷
OATP1B3 <i>K_i</i> (uM)	0.088	Pahwa S. et al ¹⁷
OATP2B1 <i>K_i</i> (uM)	0.023	Optimized from Riha J. et al ¹⁸
NTCP <i>K_i</i> (uM)	187.65	Prueksaritanont T. et al ¹⁵
CYP2C8 <i>K_i</i> (uM)	24.5	Kajosaari LI. et al ¹⁹
CYP3A4 <i>K_i</i> (uM)	15	Kajosaari LI. et al ¹⁹

BCRP, breast cancer resistance protein; B/P, blood-to-plasma ratio; *CL_{int}*, intrinsic clearance; *CL_{PD}*, passive diffusion clearance; *CL_R*, renal clearance; CYP, cytochrome P450; *Fu*, fraction unbound in plasma; *Fu_{ew}*, fraction unbound in extracellular water; *Fu_{gut}*, fraction unbound in gut; *Fu_{iw}*, fraction unbound in intracellular water; HLM,

human liver microsomes; J_{max} , maximum transporter uptake rate; K_i , inhibition constant; K_m , Michaelis-Menten constant; K_p , tissue-to-plasma partition coefficient; LogP, octanol/water partition coefficient; MRP, multidrug resistance protein; NTCP, sodium taurocholate cotransporting polypeptide; OATP, organic anion transporting polypeptide; PBPK, physiologically based pharmacokinetic; $P_{eff,man}$, effective human jejunal permeability; P-gp, P-glycoprotein; pKa, acid dissociation constant; Ref, study reference; SD, single-dose; V_{ss} , volume of distribution at steady state.

References:

1. Liu L. *et al.* 2010. Radiosynthesis and bioimaging of the tuberculosis chemotherapeutics isoniazid, rifampicin and pyrazinamide in baboons. *J Med Chem.* **53**, 2882-2889 (2010).
2. Stephen KW. *et al.* Factors determining the passage of drugs from blood into saliva. *Br J Clin Pharmacol.* **9**, 51-55 (1980).
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Supplementary Table S4. Simcyp PBPK model input parameters and source references for cyclosporine (SV-Cyclosporine_Neoral).

Parameter	Value	Method/Ref
Molecular weight (g/mol)	1202	PubChem
Log P	2.96	Meta-analysis
Compound type	Neutral	
B/P	1.62	Legg B. et al ¹
F_u	0.0365	Legg B. et al ²
Main binding protein	Human serum albumin	Akhlaghi F. et al ³
F_a	1	Assumed
k_a	1	Min D. et al ⁴ Stein CM. et al ⁵
Lag time (h)	1	Optimized to match observed data from Stein CM. et al ⁵
F_{ugut}	1	Assumed
Q_{gut} (L/h)	11.05	Predicted from scaling Caco-2 data (apical pH:basolateral pH 6.5:7.4)
Distribution model	Full PBPK model	File input
V_{ss} (L/kg)	Predicted	Method 2
CYP3A4 pathway M1 V_{max} (pmol/min/pmol)	1.7081	A retrograde approach was used to calculate the CYP3A4/5 intrinsic clearance of cyclosporine to generate metabolites, M1 (AM9), M17 (AM1), and M21(AM4N). The clearance value for each metabolic pathway was assigned based on the total CYP3A4/5 intrinsic clearance and f_m value for each pathway estimated using in vitro data (Aoyama T. et al ⁶). A 'global' K_m was calculated based on enzyme kinetic data reported by Kelly PA. et al ⁷ , which was corrected by a F_{umic} of 0.604 (Simcyp Prediction) and further used to derive the V_{max} for each metabolic pathway.
CYP3A4 pathway M1 K_m (uM)	3.02	
CYP3A4 pathway M17 V_{max} (pmol/min/pmol)	1.8395	
CYP3A4 pathway M17 K_m (uM)	3.02	
CYP3A4 pathway M21 V_{max} (pmol/min/pmol)	1.6424	
CYP3A4 pathway M21 K_m (uM)	3.02	
CYP3A5 pathway M17 V_{max} (pmol/min/pmol)	9.061	
CYP3A5 pathway M17 K_m (uM)	3.02	
Active uptake into hepatocyte	1.534	
CL_{int} (Bile) (uL/min/10 ⁶ cells)	0.69	
		Min D. et al ⁴ Ducharme MP. et al ⁸ Venkataramanan R. et al ⁹
CL_R (L/h)	0.029	Min D. et al ⁴ Ducharme MP. et al ⁸ Novartis ¹⁰
CYP3A4 K_i (uM)	0.89	Kajosaari LI. et al ¹¹
P-gp K_i (uM): Gut & Liver	0.02	Rao US. et al ¹²
BCRP K_i (uM): Gut & Liver	0.5	Ozvegy C. et al ¹³
OATP1B1 K_i (uM): Liver	0.019	Gertz M. et al ¹⁴
OATP1B3 K_i (uM): Liver	0.032	Gertz M. et al ¹⁴

BCRP, breast cancer resistance protein; B/P, blood-to-plasma ratio; CL_{int} , intrinsic clearance; CL_R , renal clearance; CYP, cytochrome P450; F_a , fraction absorbed; f_m , fraction metabolized; F_u , fraction unbound in plasma; F_{ugut} , fraction unbound in gut; F_{umic} , fraction unbound in microsomes; IV, intravenous; k_a , first-order absorption rate constant; K_i , inhibition constant; K_m , Michaelis-Menten constant; LogP, octanol/water partition coefficient; OATP, organic anion transporting polypeptide; P-gp, P-glycoprotein; Q_{gut} , gut blood flow term used in the absorption model; Ref, study reference; V_{max} , maximum metabolic rate; V_{ss} , volume of distribution at steady state.

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- Legg B. et al. Cyclosporin: Pharmacokinetics and detailed studies of plasma and erythrocyte binding during intravenous and oral administration. *Eur J Clin Pharmacol.* **34**, 451-60 (1988).
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Supplementary Table S5. Simcyp PBPK model input parameters and source references for cyclosporine metabolite M17 (AM1).

Parameter	Value	Method/Ref
Molecular weight (g/mol)	1218	ChemIDplus
Log P	2.96	Assumed same as Cyclosporine
Compound Type	Neutral	
B/P	3.03	Awni WM. et al ¹
F_u	0.031	Awni WM. et al ¹
Main binding protein	Human serum albumin	Assumed same as cyclosporine
F_{ugut}	1	Assumed
Distribution Model	Full PBPK model	
V_{ss} (L/kg)	Predicted	Method 2
Elimination Model		
CL_{IV} (L/h)	4.918	Awni WM. et al ¹ Ku YM. et al ²
CL_R (L/h)	0.11	Awni WM. et al ¹
Interaction Parameters		
Transporter	SLCO1B1 (OATP1B1)	
Organ	Liver	
K_i (μ M)	0.093	Gertz M. et al ³
Transporter	SLCO1B3 (OATP1B3)	
Organ	Liver	
K_i (μ M)	0.059	Gertz M. et al ³

B/P, blood-to-plasma ratio; CL_{IV} , intravenous clearance; CL_R , renal clearance; F_u , fraction unbound in plasma; F_{ugut} , fraction unbound in gut; K_i , inhibition constant; LogP, octanol/water partition coefficient; OATP, organic anion transporting polypeptide; PBPK, physiologically based pharmacokinetic; Ref, study reference; SLCO, solute carrier organic anion transporter; V_{ss} , volume of distribution at steady state.

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Supplementary Table S6. Summary of predicted and observed zavegepant exposure parameters for model development

		100 mg oral zavegepant					
		C_{max} (ng/mL)		AUC_{0-inf} (ng/mL·h)			
Simulated							
Geometric mean		8.94		36.85			
CV%		77		98			
Observed							
Geometric mean		11.12		42.75			
CV%		79		72			
Ratio							
Pred:Obs		0.80		0.86			
		50 mg oral zavegepant		50 mg oral zavegepant + 200 mg itraconazole QD		GMR	
		AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf}	C_{max}
Simulated							
Geometric mean		18.23	5.35	34.59	8.38	1.90	1.57
90% CI		(16.49, 20.16)	(4.95, 5.77)	(31.45, 38.04)	(7.82, 8.97)	(1.85, 1.95)	(1.53, 1.60)
Observed							
Geometric mean		25.18	7.24	39.92	12.81	1.59 ^a	1.77
90% CI		-	-	-	-	(1.16, 2.17)	(1.08, 2.89)
Ratio							
Pred:Obs		0.72	0.74	0.87	0.65	1.20	0.89
		100 mg oral zavegepant		100 mg oral zavegepant + 600 mg rifampin QD		GMR	
		AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf}	C_{max}
Simulated							
Geometric mean		38.80	10.93	90.72	21.25	2.34	1.94
90% CI		(35.12, 42.87)	(10.16, 11.76)	(83.74, 98.27)	(19.97, 22.61)	(2.24, 2.46)	(1.87, 2.02)
Observed							
Geometric mean		39.04	11.12	90.94	24.12	2.33	2.17
90% CI		-	-	-	-	(1.80, 3.02)	(1.60, 2.94)
Ratio							
Pred:Obs		0.99	0.98	1.00	0.88	1.01	0.90

^aReported AUC_{0-last} ratio is 1.74 (90% CI = 1.28, 2.35).

AUC_{0-inf} , area under the curve from time 0 to infinity; CI, confidence interval; C_{max} , maximal drug concentration; CV, coefficient of variation; GMR, geometric mean ratio; Obs, observed; Pred, predicted; QD, once daily.

Supplementary Table S7. Sensitivity analysis of intestinal and hepatic efflux mediated by P-gp and other unknown efflux transporters on the impact of simulated itraconazole DDI effects^a

P-gp vs other efflux	<i>AUC</i>_{0-last} ratio	<i>C</i>_{max} ratio	<i>Fa</i> increase
Intestinal 1:0	6.5	3.4	141%
Intestinal 1:2	3.1	1.7	21%
Intestinal 1:5	2.8	1.5	9%
Intestinal and hepatic 1:2	1.67	1.40	21%
Observed	1.74	1.77	-

^aSimulations considered the inhibitory effect on cytochrome P450 3A4 (CYP3A4) and P-gp by itraconazole and OH-itraconazole.

*AUC*_{0-last}, area under the curve from 0 to time of last measured concentration; *C*_{max}, maximal drug concentration; DDI, drug–drug interaction; *Fa*, fraction absorbed; P-gp, P-glycoprotein.

Supplementary Table S8. Sensitivity analysis of relative contribution between active (OATP1B3) and passive transport in the liver on the impact of simulated rifampin effects

Active vs passive	$CL_{int,OATP1B3}$ and CL_{PD} ($\mu\text{L}/\text{min}/\text{millions}$)	AUC ratio	C_{max} ratio
50:50	20 and 19.4	1.25	1.35
90:10	36 and 4	3.30	2.41
80:20 ^a	32 and 8	2.34	2.00
80:20 ^b	28 and 7	2.32	1.94
Observed	-	2.33	2.17

^aPredicted CL/F = 2950 L/h.

^bDue to a slight over-prediction of CL/F , the overall hepatic uptake clearance is reduced from 40 to 35 $\mu\text{L}/\text{min}/\text{millions}$, which leads to a predicted CL/F of 2700 L/h. This was used in the final model.

AUC, area under the curve; CL/F , apparent clearance; CL_{PD} , passive diffusional clearance; $CL_{int,OATP1B3}$, intrinsic biliary clearance via OATP1B3 uptake; C_{max} , maximal drug concentration; OATP1B3, organic anion transporting polypeptide 1B3.

Supplementary Table S9. Sensitivity analysis of percent of dose inhaled vs swallowed following intranasal dosing of 20 mg zavegepant once daily for 14 days

Percent inhaled vs swallowed	<i>AUC</i>_{0-24,ss} (ng/mL·h)	<i>C</i>_{max,ss} (ng/mL)
90:10	831	270
80:20	740	240
20:80	192	62
10:90	101	32
Observed	95.0	40.9

*AUC*_{0-24,ss}, area under the curve from time 0 to 24 hours at steady state; *C*_{max,ss}, maximal drug concentration at steady state.

Supplementary Table S10. Summary of predicted and observed zavegepant exposure parameters following oral administration in simulations conducted for model validation

	Day 1		Day 10	
	C_{max} (ng/mL)	AUC_{0-24} (ng/mL·h)	C_{max} (ng/mL)	$AUC_{0-24,ss}$ (ng/mL·h)
50 mg oral zavegepant				
Predicted				
Geometric mean	5.32	16.69	5.37	17.41
CV%	57	80	58	82
Observed				
Geometric mean	5.29	18.53	5.72	23.42
CV%	58	57	135	71
Ratio				
Pred:Obs	1.00	0.90	0.94	0.74
100 mg oral zavegepant				
Predicted				
Geometric mean	10.91	34.35	11.02	35.89
CV%	67	95	68	97
Observed				
Geometric mean	12.1	41.71	9.01	38.46
CV%	147	103	30	28
Ratio				
Pred:Obs	0.90	0.82	1.22	0.93
200 mg oral zavegepant				
Predicted				
Geometric mean	25.21	78.08	25.46	81.57
CV%	64	92	65	95
Observed				
Geometric mean	21.33	84.56	23.61	91.44
CV%	39	37	167	77
Ratio				
Pred:Obs	1.18	0.92	1.08	0.89
300 mg oral zavegepant				
Predicted				
Geometric mean	33.42	102.44	33.74	106.97
CV%	68	96	68	98
Observed				
Geometric mean	24.9	109.82	38.88	147.49
CV%	79	65	61	52
Ratio				
Pred:Obs	1.34	0.93	0.87	0.73

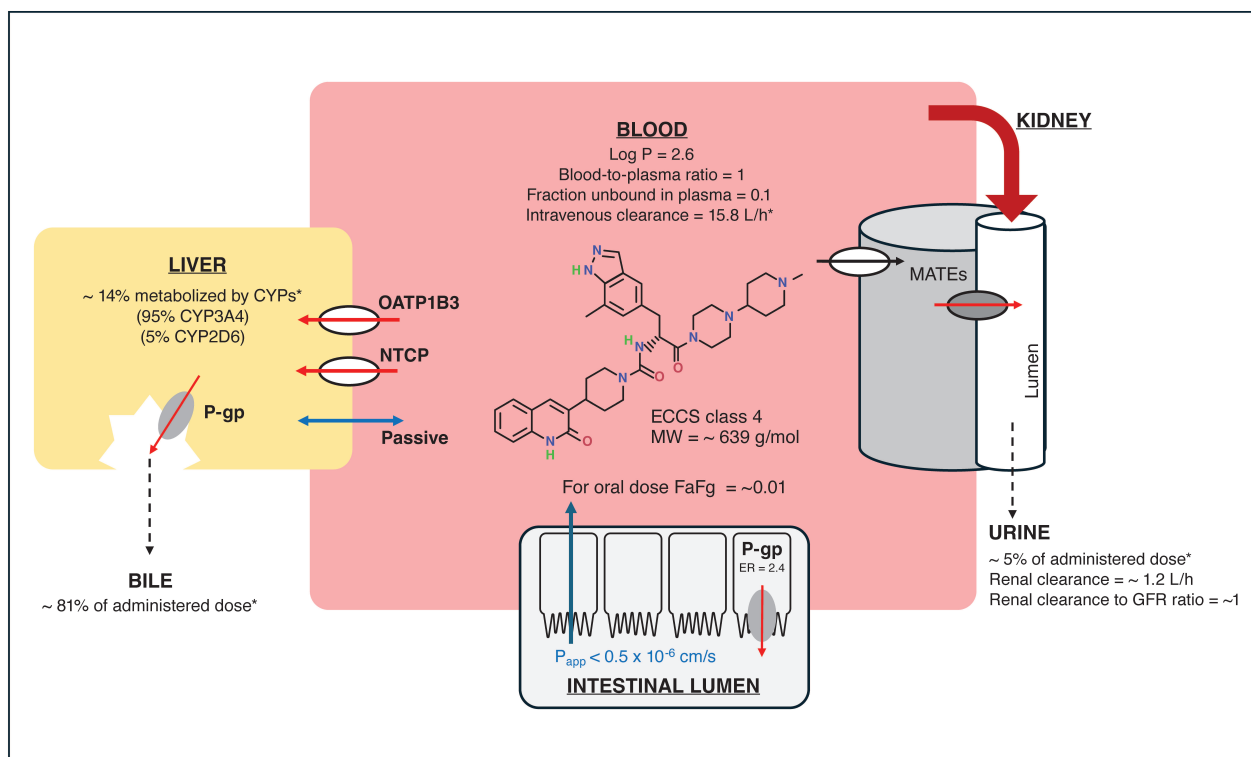
AUC_{0-24} , area under the curve from time 0 to 24 hours; C_{max} , maximal drug concentration; CV, coefficient of variation; Obs, observed; Pred, predicted.

Supplementary Table S11. Summary of predicted and observed zavegepant exposure parameters following intranasal administration for model validation

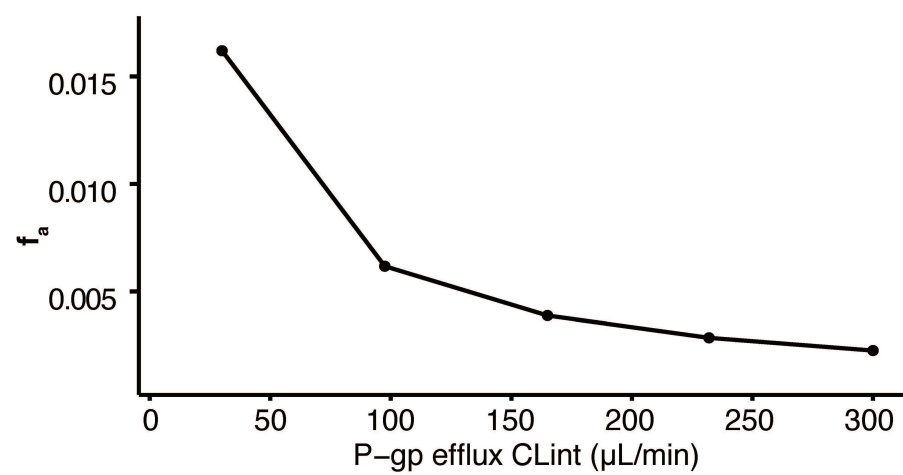
Following intranasal administration for model validation						
	5 mg IN zavegepant					
	Day 1			Day 14		
	C_{max} (ng/mL)	AUC_{0-24} (ng/mL·h)		C_{max} (ng/mL)	$AUC_{0-24,ss}$ (ng/mL·h)	
Predicted						
Geometric mean	7.81	23.81		7.90	24.97	
CV%	43	56		43	59	
Observed						
Geometric mean	11.37	27.34		7.58	23.93	
CV%	77	63		64	55	
Ratio						
Pred:Obs	0.69	0.87		1.04	1.04	
	10 mg IN zavegepant					
	Day 1			Day 14		
	C_{max} (ng/mL)	AUC_{0-24} (ng/mL·h)		C_{max} (ng/mL)	$AUC_{0-24,ss}$ (ng/mL·h)	
Predicted						
Geometric mean	15.08	46.09		15.23	48.30	
CV%	44	57		45	59	
Observed						
Geometric mean	16.31	32.10		12.98	36.49	
CV%	72	78		47	44	
Ratio						
Pred:Obs	0.92	1.44		1.17	1.32	
	10 mg IN zavegepant		10 mg IN zavegepant +200 mg itraconazole QD		GMR	
	AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf} (ng/mL·h)	C_{max} (ng/mL)	AUC_{0-inf}	C_{max}
Predicted						
Geometric mean	47.81	14.87	65.02	16.95	1.36	1.14
90% CI	(44.96, 50.85)	(14.21, 15.57)	(61.15, 69.14)	(16.22, 17.71)	(1.34, 1.37)	(1.13, 1.15)
Observed						
Geometric mean	48.01	23.93	48.81	20.99	1.02	0.88
90% CI	-	-	-	-	(0.82, 1.26)	(0.64, 1.20)
Ratio						
Pred:Obs	1.00	0.62	1.33	0.81	1.34	1.30

AUC_{0-inf} , area under the curve from time 0 to infinity; CI, confidence interval; C_{max} , maximal drug concentration; CV, coefficient of variation; GMR, geometric mean ratio; IN, intranasal; Obs; observed; Pred, predicted; ss, steady state.

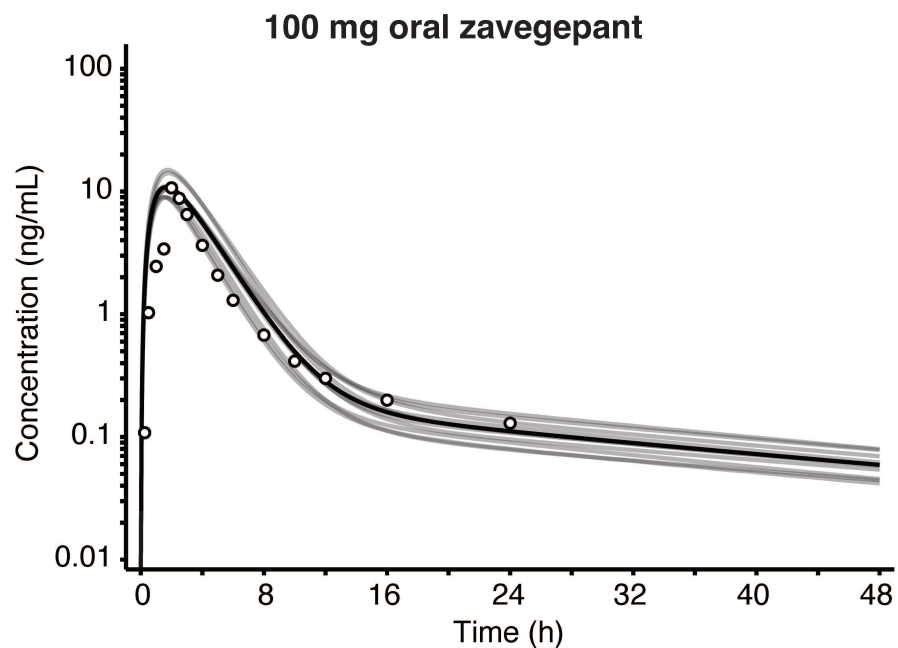
Supplementary Figure S1. Summary of zavegepant human absorption, distribution, metabolism, and excretion. *These values in the model were based on preliminary data from a study examining 5 mg intravenous [^{14}C]-zavegepant absorption, distribution, metabolism, and excretion. Bile = 80.5% of dose excreted unchanged, urine = 5% of dose excreted unchanged, CYP metabolism = remaining portion (100% – 80.5% – 5% = 14.5%), and intravenous clearance = 15.8 L/h. However, final data from the intravenous study were as follows: Bile = 80.5% of dose excreted unchanged, urine = 6.2% of dose excreted unchanged, CYP metabolism = remaining portion (100% – 80.5% – 6.4% = 13.1%), and intravenous clearance = 13.2 L/h. These minor differences are not expected to impact the conclusion of the analysis. CYP, cytochrome P450; ER, efflux ratio; ECCS, extended clearance classification system; FaF_g , fraction absorbed fraction not metabolized by the gut; GFR, glomerular filtration rate; MATE, multidrug and toxin extrusion protein; MW, molecular weight; OATP1B3, organic anion transporting polypeptide 1B3; P_{app} , apparent permeability coefficient; P-gp, P-glycoprotein; NTCP, Na $^{+}$ /taurocholate co-transporting polypeptide.



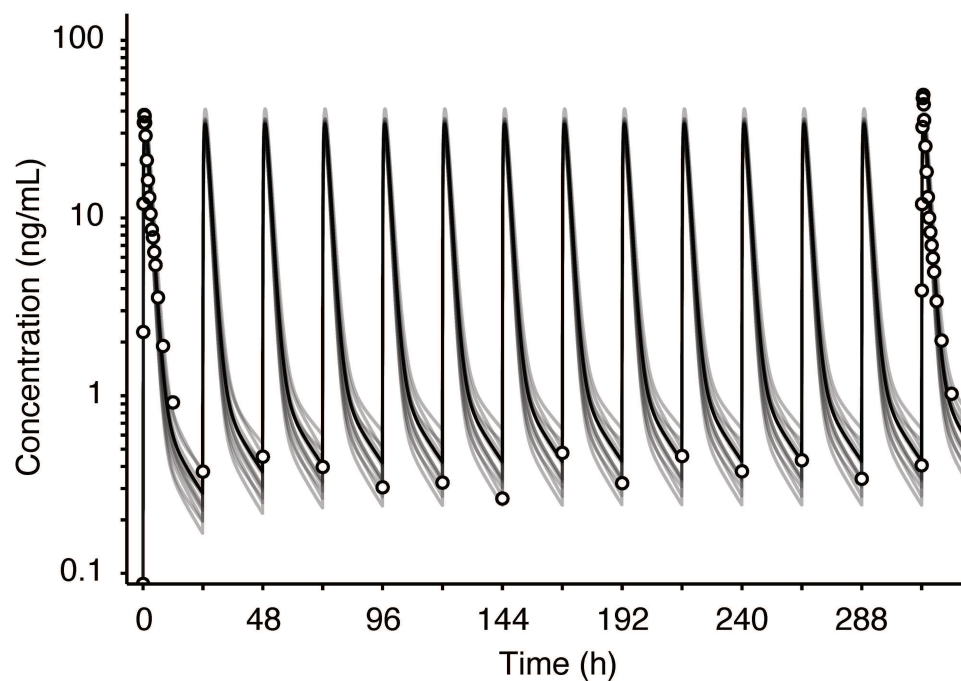
Supplementary Figure S2. Sensitivity analysis of intestinal P-gp efflux CL_{int} . CL_{int} , intrinsic transporter clearance; F_a , fraction absorbed; P-gp, P-glycoprotein.



Supplementary Figure S3. Simulation of a single oral dose of 100 mg zavegepant (model development). Gray lines represent mean zavegepant concentrations of the simulated individual trials, and black lines represent mean concentration for the simulated population ($n = 150$). Circles represent observed zavegepant concentrations ($n = 15$).

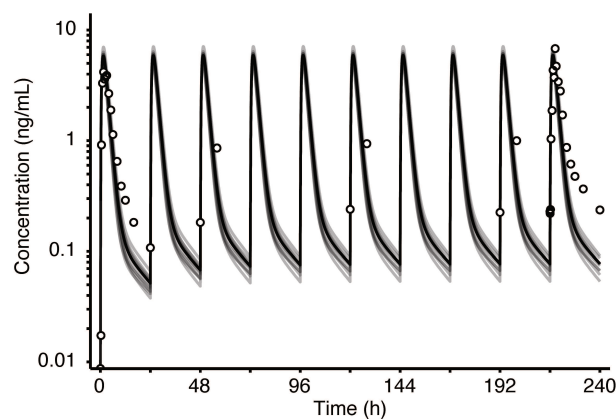


Supplementary Figure S4. Simulations of single and multiple IN doses of 20 mg zavegepant (model development). Lines represent simulated zavegepant concentrations, and circles represent observed concentrations (mean of $n = 9$ individuals). Gray lines represent mean zavegepant concentrations of simulated individual trials, and solid black lines represent the mean concentrations for the simulated population ($n = 90$). IN, intranasal.

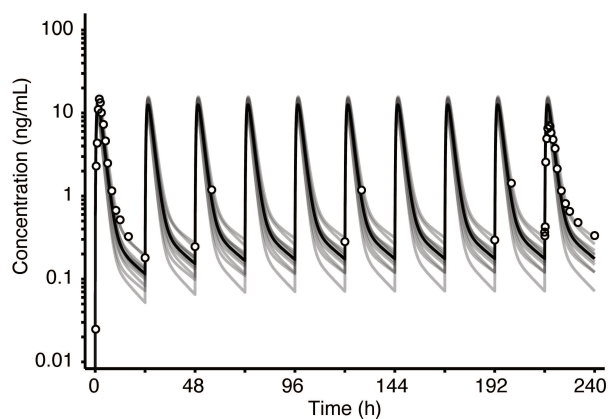


Supplementary Figure S5. Simulations of single and multiple oral doses of 50 mg (A), 100 mg (B), 200 mg (C), and 300 mg (D) zavegepant (model validation). Lines represent simulated zavegepant concentrations, and circles represent observed concentrations (mean of $n = 12$ individuals). Gray lines represent mean zavegepant concentrations of simulated individual trials, and solid black lines represent the mean concentrations for the simulated population ($n = 120$).

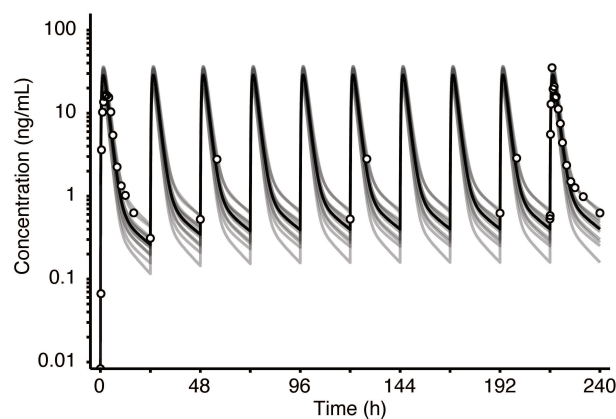
(A) 50 mg oral zavegepant



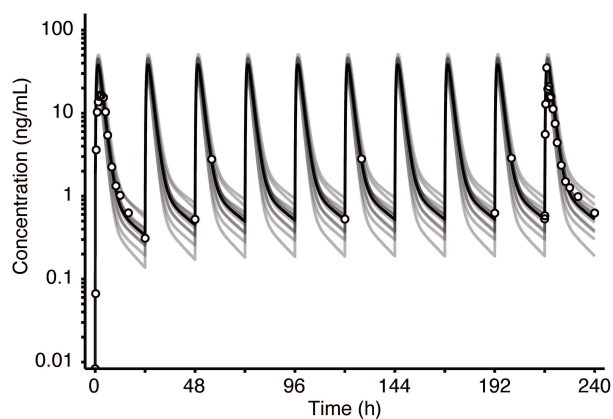
(B) 100 mg oral zavegepant



(C) 200 mg oral zavegepant

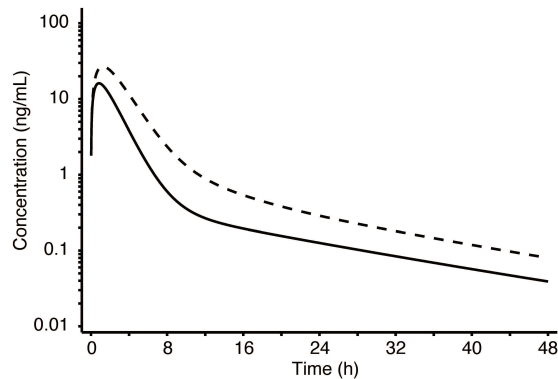


(D) 300 mg oral zavegepant

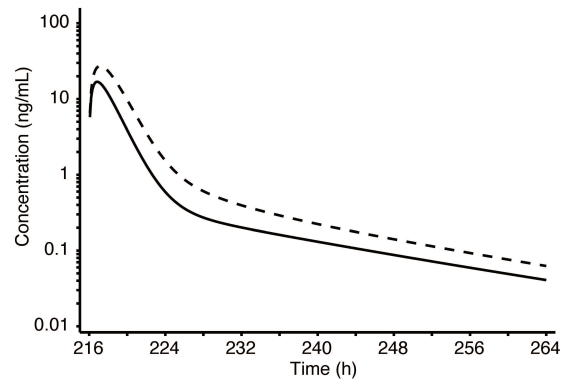


Supplementary Figure S6. Simulations of a single IN dose of 10 mg zavegepant administered alone and with SD rifampin (A), MD rifampin (B), SD cyclosporine A (C), MD carbamazepine (D-E), or MD efavirenz (F) conducted during model application. Lines represent mean zavegepant concentration for the simulated population ($n = 100$) in the absence (solid lines) or presence (dashed lines) of the specified concomitant medication. Zavegepant was administered with SD 600 mg rifampin (A), on the 10th day of 10 days of dosing with 600 mg rifampin QD (B), with SD 600 mg cyclosporine A (C), on the 10th day of 11 days of dosing with 300 mg carbamazepine BID (D-E), or on the 10th day of 11 days of dosing with 600 mg efavirenz QD (F). For carbamazepine, simulations in (D) incorporated CYP3A4 induction only, and simulations in (E) incorporated induction of both CYP3A4 and P-gp. BID, twice a day; CYP3A4, cytochrome P450 3A4; IN, intranasal; MD, multiple dose; P-glycoprotein; QD, once daily; SD, single dose.

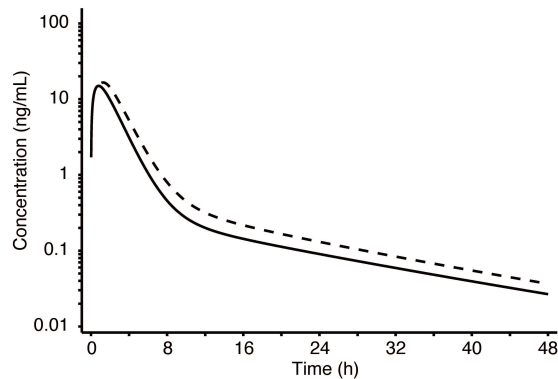
(A) 10 mg IN zavegepant $\pm$ 600 mg rifampin SD



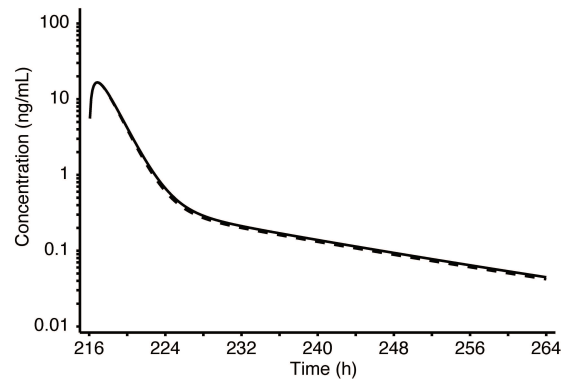
(B) 10 mg IN zavegepant $\pm$ 600 mg rifampin QD



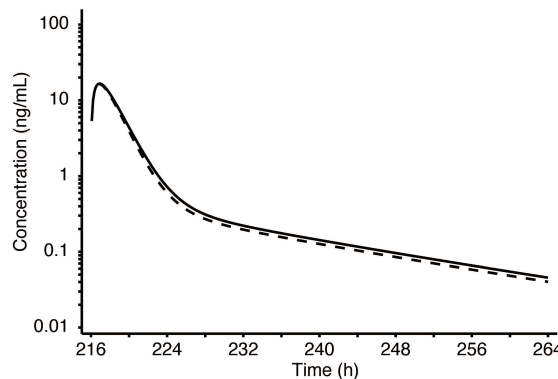
(C) 10 mg IN zavegepant $\pm$ 600 mg cyclosporine A SD



(D) 10 mg IN zavegepant $\pm$ 300 mg Carbamazepine BID (CYP3A4 induction only)



(E) 10 mg IN zavegepant $\pm$ 300 mg carbamazepine BID (CYP3A4 and P-gp induction)



(F) 10 mg IN zavegepant $\pm$ 600 mg efavirenz QD

